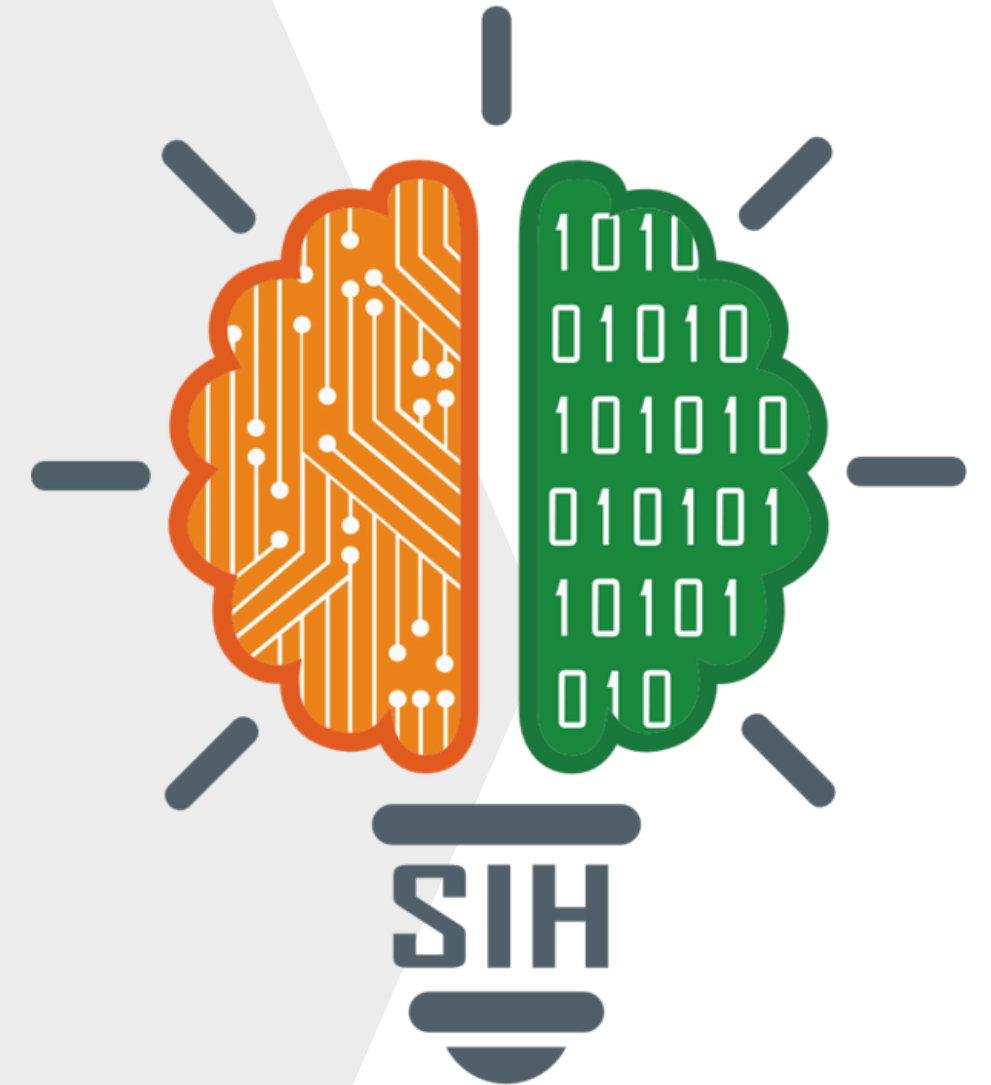
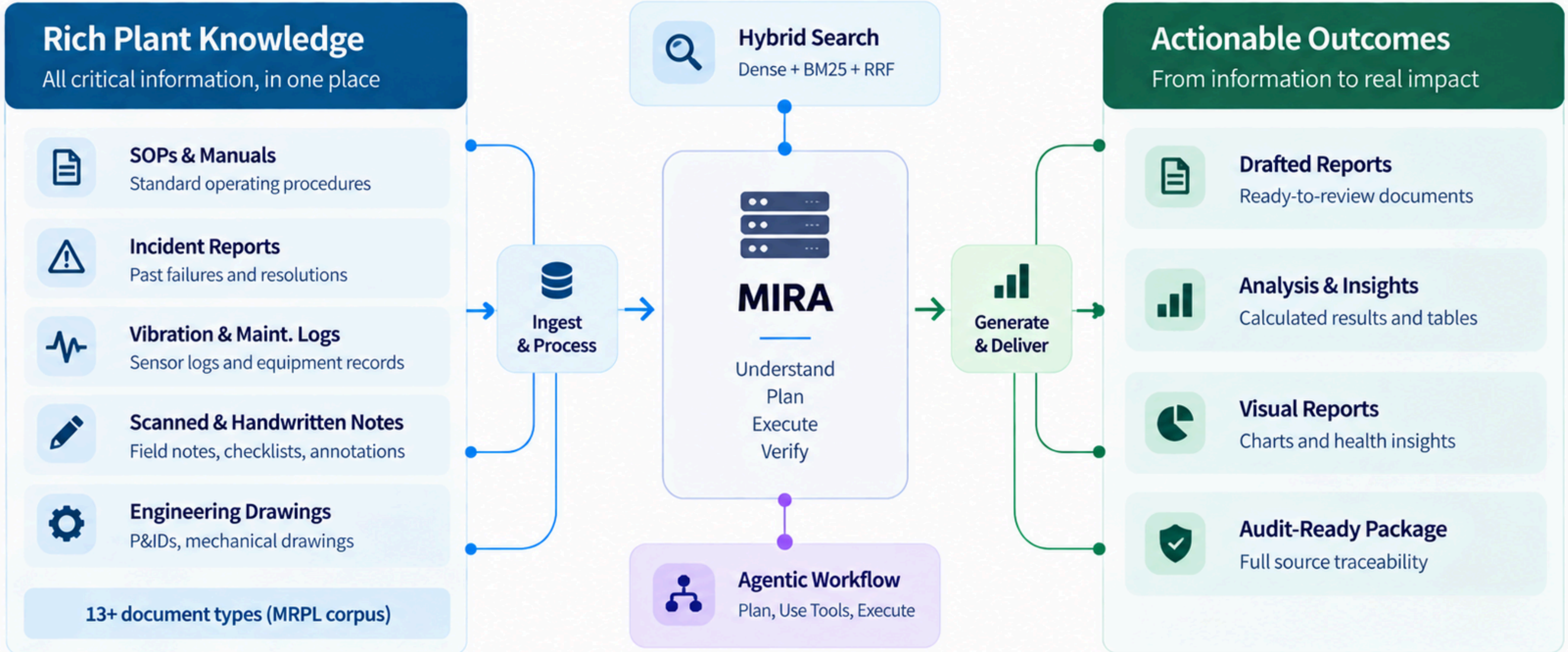


TITLE PAGE

- Problem Statement ID – 26117
- Problem Statement Title- Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work
- Theme- Smart Automation
- PS Category- Software
- Team ID - S155
- Team Name - LocalHost





TECH STACK & HARDWARE

Tools powering the solution



Python

Core development, agent logic



Ollama (Local LLM)

Open-weight, offline inference



Tesseract OCR

Scan & handwritten text extraction



Hybrid Search / RRF

Semantic + keyword + tag search



Streamlit

Web interface (UI)



Custom Agent (No LangChain)

Domain-specific tools & logic



Excel / Python Libraries

Analysis & report generation



NVIDIA RTX 3050 / 5060

Runs locally on laptop

TECHNICAL APPROACH

DOCUMENT PROCESSING PIPELINE

From raw plant documents to actionable outputs



FLAGSHIP SCENARIO



KEY SYSTEM COMPONENTS

Core modules of the solution



KB Index

Hybrid dense + BM25 + tag match



Sandbox

Isolated code execution



Netguard

Blocks & proves no internet access



Audit Log

Tamper-proof traceability

METHODOLOGY & IMPLEMENTATION

From development to deployment



Build document ingestion & OCR pipeline



Implement hybrid retrieval (dense + BM25 + RRF)



Develop agent with tool calling and local execution



Integrate human approval gate



Add audit logging and provenance tracking



Iteratively test on real MRPL documents and refine

FEASIBILITY ANALYSIS

Is it practical and doable?



Technical Feasibility

Runs on commodity hardware (e.g., RTX 3050/5060, 8–16 GB RAM).



Data Feasibility

MRPL documents (SOPs, logs, scans) are available in multiple formats.



Operational Feasibility

Integrates with existing plant workflows and can be deployed locally (air-gapped).



Economic Feasibility

Cost-effective using open-source tools (Ollama, Tesseract, LangChain, Streamlit).

CHALLENGES & RISKS

What could go wrong?



Large & Diverse Documents

Scanned, handwritten and multi-format files may reduce OCR accuracy.



Model Limitations

Local LLMs (e.g. Ollama) may generate incorrect or incomplete outputs (hallucinations).



Infrastructure Constraints

Air-gapped environment limits to external APIs and internet resources.



Team & Development Risks

Limited development time and need for synchronization across team members.

MITIGATION STRATEGIES

How we will overcome them?



Improve Document Processing

Use advanced OCR (Tesseract), structure-aware chunking and tag repair against equipment registers.



Domain-Tuned Retrieval

Hybrid search (dense + BM25 + RRF) with MRPL terminology to reduce hallucinations.



Fully Local & Secure Setup

Run models and tools on local GPU infrastructure; no internet dependency.

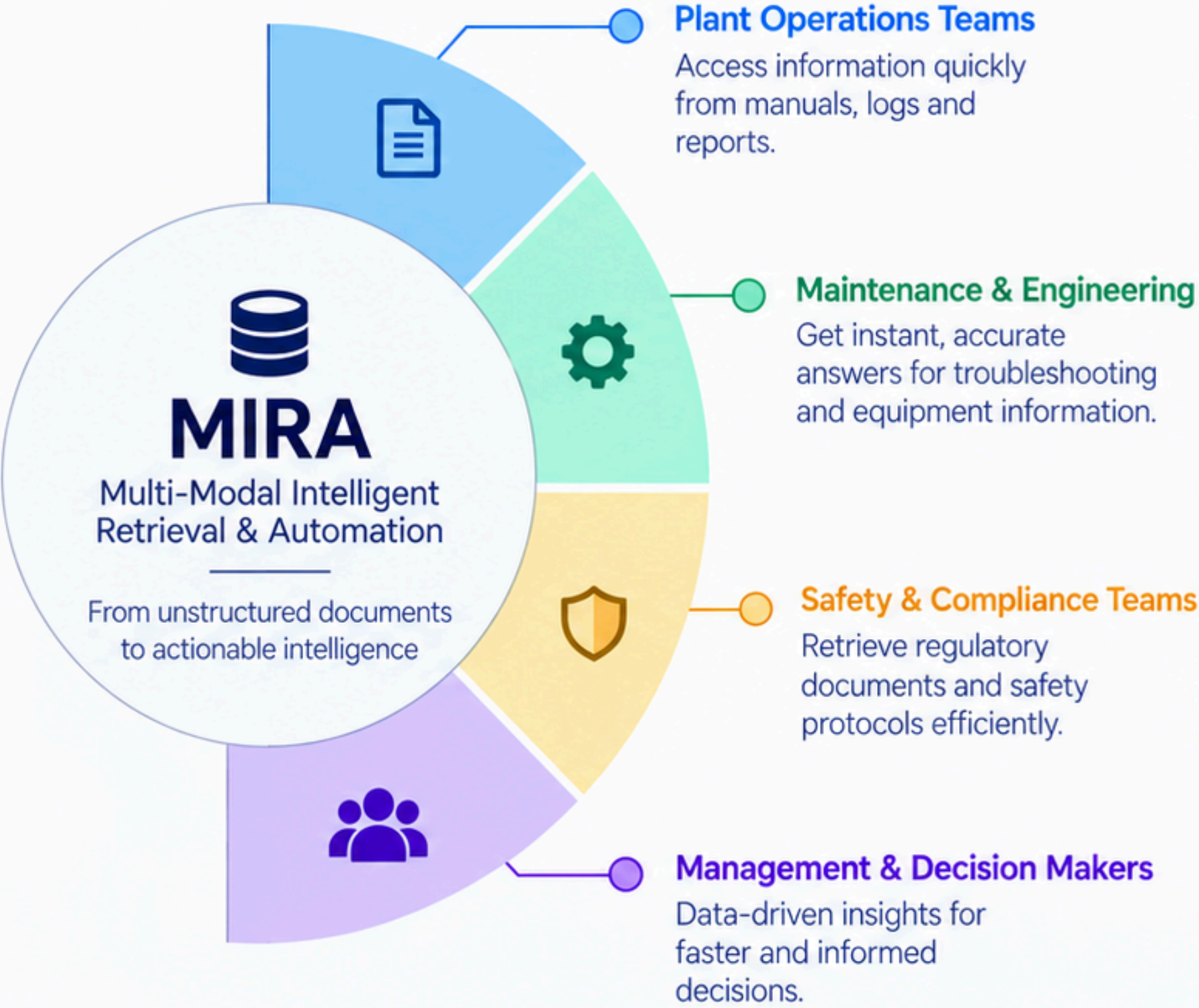


Iterative Testing & Validation

Continuous testing on real MRPL documents, human-in-the-loop review and feedback-driven improvements.

LocalHost

TARGET AUDIENCE



BROADER IMPACT

Long-term value beyond the plant

Impact Area	Benefits	Examples
 Operational Efficiency	Faster access to critical information and reduced document search time.	Quick retrieval of SOPs, manuals and maintenance guides.
 Economic Impact	Lower operational cost through automation and reduced downtime.	Less manual effort, faster problem resolution and better resource use.
 Compliance & Safety	Improved adherence to regulatory standards with accurate information access.	Faster audit readiness and reduced compliance risks.
 Social Impact	Empowers employees with easy access to knowledge and training.	Better workforce productivity and skill development.
 Scalability	Adaptable to other plants and industries with minimal customization.	Deployment across multiple MRPL units and similar sectors.



Reciprocal Rank Fusion

Reciprocal Rank Fusion outperforms Condorcet and individual rank learning methods. SIGIR 2009.

DOI: 10.1145/1571941.1572114

<https://doi.org/10.1145/1571941.1572114>



Tesseract OCR Engine

OCR engine — Tesseract OCR.

<https://github.com/tesseract-ocr/tesseract>



Built on Proven Research

Leveraging
state-of-the-art models,
established methods and
real-world data.



Qwen2.5 Technical Report

Language model — Qwen2.5
Technical Report (Yang, A., et al., 2024).
arXiv:2412.15115

<https://arxiv.org/abs/2412.15115>